

Learning Seminar on Statement of Arithmetic Product Formula Conjecture

Fall 2024/2025

Time: Monday 2:00pm-4:00pm ET time Place: Room 2-255 at MIT building 2

In this seminar, we want to understand statement of arithmetic product formula conjecture, cf. [Liu11, L-L21].

- 1. Classical theta correspondence, Siegel-Weil formula, Doubling method.** Reference: [Li24]2.1-2.3, 2.4 before Rallis inner product formula.
- 2. Rallis inner product formula and central derivatives.** References: [Li24]2.4-2.5, [Li92]for proof, [Liu11]2D.
- 3. Arithmetic and geometric theta series.** Unitary Shimura variety, special cycles, arithmetic theta function, geometric theta function. References: [Li24]3.1-3.5.
- 4. Geometric Siegel-Weil formula and geometric inner product formula.** Modularity of generating series, geometric Siegel-Weil and geometric inner product formula, sketch of proof if time permits. References: [Li24]3.6-3.7, [Dun22]section2.2.
- 5. Arithmetic Chow groups and arithmetic intersection theory** Define Green currents, arithmetic Chow group(3.1-3.3) and cup product pairing(thm 4.2.3,4.2.10)). References: [G-S90]3.1-3.3., 4.2..
- 6. Integral models and Arakelov theta series.** Integral models of unitary Shimura variety, special cycles, Arakelov theta function, and modularity. References: [Li24]section4.1-4.3, [Liu11]3B, and [BHKRY].
- 7. Arithmetic Siegel-Weil formula and arithmetic inner product formula.** Define both sides of the Inner product formula and arithmetic Siegel-Weil formula. References: [Li24]4.4-4.5.
- 8. Beilinson-Bloch height pairing and local indices.** Define the Beilinson-Bloch height pairing under suitable conditions, and illustrate how to calculate them. References: [L-L21]Appendix B and section6, [Bei87].
- 9. Sketch of proof of arithmetic inner product formula.** Sketch proof of arithmetic inner product formula and applications. References are [L-L21]section6-11.

References

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- [BHKRY] J. Bruinier, B. Howard, S. Kudla, M. Rapoport and T. Yang, Modularity of generating series of divisors on unitary Shimura varieties. *Astérisque, Diviseurs arithmétiques sur les variétés orthogonales et unitaires de Shimura*(2020), no.421, 7-125.
- [Dun22] N. C. Dũng, Geometric Pullback Formula for Unitary Shimura Varieties. Thesis (Ph.D.)-Columbia University. ProQuest LLC, Ann Arbor, MI, 2022. 68 pp.
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- [Li24] C. Li, Geometric and arithmetic theta correspondences. <https://www.math.columbia.edu/~chaoli/IHES.pdf>.
- [L-L21] C. Li and Y. Liu, Chow groups and L -derivatives of automorphic motives for unitary groups. Ann. of Math. (2)194(2021), no.3, 817-901.
- [Li92] J.-S. Li, Nonvanishing theorems for the cohomology of certain arithmetic quotients. J. Reine Angew. Math.428(1992), 177-217.
- [Liu11] Y. Liu, Arithmetic theta lifting and L -derivatives for unitary groups, I. Algebra Number Theory5(2011), no.7, 849-921.
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